

JOURNAL ARTICLE

Efficient-EF-OFL: An Entropy-Based Filtering and Ordinal Focal Loss Approach with EfficientNet for Prostate Lesion Classification

Woshington V S Rodrigues¹ | Armando Luz Borges² | José D. L. Araújo³ | Antonio O. C. Filho⁴

¹Postgraduate Program Doctorate in Computer Science, UFMA/UFPI Association, Ministro Petrônio Portella University Campus, PI, Brazil

²Graduate, UFMA/UFPI Association, Ministro Petrônio Portella University Campus, PI, Brazil

³Postgraduate Program Doctorate in Computer Science, UFMA/UFPI Association, Ministro Petrônio Portella University Campus, PI, Brazil

⁴Postgraduate Program Doctorate in Computer Science, UFMA/UFPI Association, Ministro Petrônio Portella University Campus, PI, Brazil

Correspondence

Corresponding author Woshington Valdeci.
Email: woshington@ifpi.edu.br

Abstract

Automated ISUP grade classification in prostate biopsy whole-slide images remains challenging due to high morphological variability, label noise, and inter-institutional heterogeneity. This study proposes an efficient pipeline combining entropy-based noise filtering, systematic patch extraction, ordinal regression, and focal loss to enhance robustness in learning from large-scale data. Variants of the EfficientNet architecture (B0, B3, and B7) were trained using a unified strategy, and their outputs were aggregated through ensemble methods. Experiments were conducted on the multicentric PANDA dataset, comprising 10,616 WSIs. The best individual model (EfficientNet-B0 with ordinal focal loss) achieved a QWK of 0.856, while deeper variants did not yield further improvements. The simple-average ensemble surpassed all single models, reaching a QWK of 0.879 and producing the most clinically coherent confusion matrix, with misclassifications largely restricted to adjacent ISUP grades. These results demonstrate that combining ordinal modeling, noise filtering, and lightweight architectures provides an advantageous balance between accuracy, stability, and computational efficiency. Although competitive with state-of-the-art approaches, the method requires validation on external datasets to further assess generalizability. Future directions include specialized models for intermediate grades and the exploration of transformer-based architectures.

KEYWORDS

Prostate cancer; ISUP grading; EfficientNet; Ordinal Regression; Ensemble Learning; Noise Filtering

1 | INTRODUCTION

Prostate cancer is one of the most common malignant neoplasms in men, representing a significant challenge for public health¹. Tumor grading plays a central role in defining clinical management and prognosis. The Gleason system, based on the sum of the two most prevalent histological patterns (Gleason score), has evolved into the ISUP grading system (0–5), which provides improved prognostic stratification and is widely used in clinical practice².

The traditional histopathological evaluation of whole-slide images (WSIs) relies on manual analysis by pathologists, a process that is subject to inter- and intra-observer variability³, time-consuming, and limited by the availability of specialists. These constraints have motivated the development of automated deep learning methods.

Convolutional neural networks (CNNs) have shown outstanding performance in medical image analysis⁴, including the automated grading of Gleason patterns and ISUP grades^{5,3}. Among modern architectures, the EfficientNet family⁶ offers an excellent trade-off between accuracy and computational cost, due to its compound scaling strategy and the use of MBConv blocks with squeeze-and-excitation⁷.

In this work, we investigate the automated classification of ISUP grades by evaluating EfficientNet variants (B0, B3, and B7) on the PANDA dataset⁸. We employ transfer learning, fine-tuning strategies, advanced training techniques, and ensemble methods to enhance model robustness. The main contributions of this study include: (1) the implementation of a loss function that combines ordinal and focal components; (2) a comparison of training techniques, including noise reduction, SWA⁹, and

Abbreviations: ANA, anti-nuclear antibodies; APC, antigen-presenting cells; IRF, interferon regulatory factor.

focal loss; (3) the development of ensemble models to improve prediction stability; and (4) a systematic analysis of EfficientNet variants for ISUP classification. Performance is primarily evaluated using the quadratic weighted kappa (QWK)¹⁰, an appropriate metric for ordinal classification tasks, along with accuracy.

2 | RELATED WORKS

The application of deep learning techniques to the automated classification of prostate cancer has been extensively investigated, with significant advances particularly in the use of whole-slide images (WSIs), patch-based strategies, and multi-scale aggregation mechanisms.

³ developed a deep learning system for automated Gleason grading using a dataset of 5,759 biopsies from Radboud University Medical Center, stratified into training (4,712), validation (497), and test (535) sets. The method, based on an extended U-Net architecture with semi-supervised labeling, achieved a quadratic weighted kappa (κ_{quad}) of 0.918 on the internal test set. System validation also included an observer study, in which the model outperformed 10 out of 15 pathologists, as well as evaluation on an independent external dataset from the University Hospital of Zurich, achieving $\kappa_{quad} > 0.700$.

⁵ developed a two-stage deep learning system (InceptionV3 CNN and k-NN) for Gleason grading using data from TCGA and two medical centers in the United States. The model was trained on 112 million annotated patches extracted from 1,226 slides and validated on an independent cohort of 331 prostatectomy slides. The system achieved an accuracy of 0.70 against an expert reference standard, significantly outperforming the average accuracy of 0.61 obtained by a panel of 29 pathologists ($p = 0.002$), and demonstrated superior performance in clinical risk stratification.

Patch-based approaches have evolved rapidly. ¹¹ conducted a pilot study to develop a deep learning algorithm specifically focused on prostate biopsies using ResNet-18. The model was trained and validated on 14,803 image patches extracted from 85 WSIs from 25 patients at a single institution. The system achieved 91.5% accuracy for malignancy detection (AUC = 0.83) and 85.4% accuracy for multiclass Gleason score classification, including benign, Gleason 3, Gleason 4, and Gleason 5.

¹² proposed an approach based on the Inception-v3 CNN architecture for the detection of Gleason patterns and the automated determination of Grade Groups in prostate biopsies. The study used a dataset comprising 96 histological tissue sections from 38 patients, each with detailed pixel-level annotations. The method generated probability maps to compute biopsy composition. The system achieved 92% accuracy in malignancy detection and 90% in distinguishing between low-grade ($GP \leq 3$) and high-grade ($GP \geq 4$) patterns. In the final slide-level Grade Group classification, the model achieved a quadratic weighted kappa (κ_{quad}) of 0.70.

¹³ proposed a system for Gleason grading using a three-scale pyramidal CNN and a pixel-wise classification strategy with majority voting. The model was developed using 608 WSI images from Radboud University Medical Center, applying a balancing strategy to mitigate the stroma pattern's predominance. The approach, which integrates edge-enhancement preprocessing, achieved 94% accuracy in the final determination of Grade Groups and demonstrated computational efficiency, operating with approximately 1 million parameters.

Other studies expanded the focus to full WSIs. ¹⁴ developed a deep learning system for biopsy grading based on a U-Net architecture with a ResNeXt50 encoder, using domain-agnostic training strategies and active learning to mitigate staining variations. The model was developed using 6,670 WSIs from three different institutions and demonstrated strong generalization performance, achieving 83.1% accuracy and a quadratic weighted kappa (κ_{quad}) of 0.93 on an external dataset, while maintaining AUCs above 0.93 for clinical risk stratification across different centers.

Despite these advances, the field still faces limitations related to data scale and methodology. Many studies rely on small datasets^{11,12}, which compromises generalization. Even robust works focused on domain adaptation, such as¹⁴, place limited emphasis on architectural efficiency and on ordinal regression. Moreover, the potential of noise-filtering techniques and uncertainty-based curation remains underexplored compared to traditional classification and segmentation approaches.

In this context, the present work distinguishes itself by systematically evaluating multiple EfficientNet variants (B0, B3, and B7) for ISUP grading on the PANDA dataset, combining ordinal regression and focal loss to model the ordinal and imbalanced nature of the classes. We also explore optimized ensemble strategies that consistently outperform individual models and incorporate a Shannon entropy-based filtering step to mitigate noisy labels, an approach not explored in the studies reviewed in this work.

3 | MATERIALS AND METHODS

The methodology adopted in this study consists of a structured pipeline for the automatic classification of ISUP grades in prostate histopathological images, as illustrated in Figure 1. The PANDA dataset is first acquired and preprocessed, then split into training and test sets. Next, noise-reduction filtering is applied exclusively to the training set, with the goal of removing potentially inconclusive images that could compromise the models' learning. After this step, whole-slide images are fragmented into patches, reducing spatial dimensionality and making training computationally more efficient. In the modeling phase, different individual classifiers are trained using distinct learning strategies. Finally, an ensemble approach is used to combine the predictions of these models, yielding a more robust and stable solution.

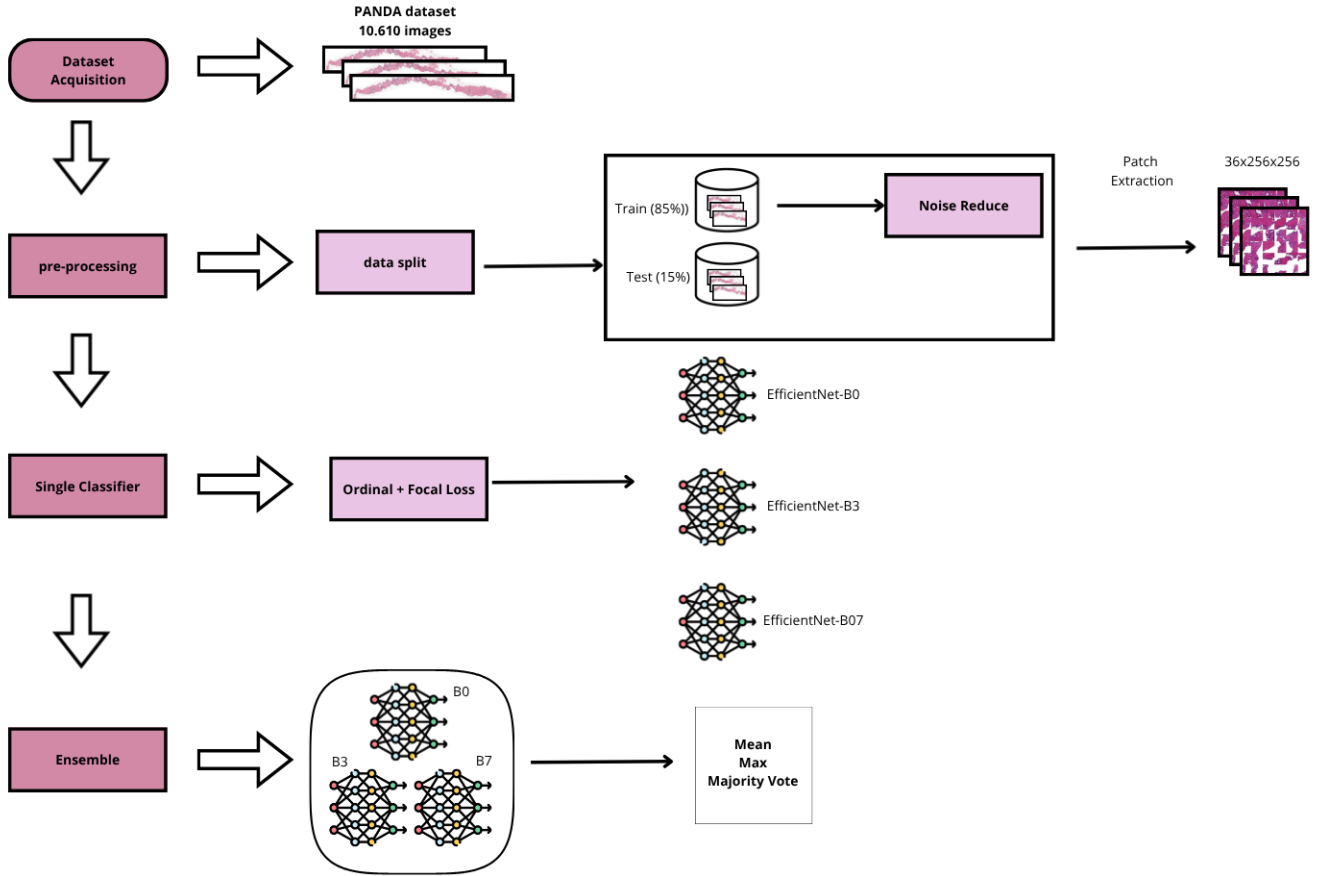


FIGURE 1 proposed method

3.1 | Dataset Acquisition

We used the Prostate Cancer Grade Assessment (PANDA) dataset⁸, which consists of 10,616 whole-slide images (WSIs) of high-resolution (20×) digitized prostate biopsies. Each image is annotated according to the ISUP grading system (0–5), where grade 0 indicates the absence of malignancy, and grades 1–5 represent increasing levels of tumor aggressiveness.

The Table 1 presents the class distribution of the dataset. A significant class imbalance is observed, with a predominance of class 0 (approximately 27%). In contrast, the classes associated with more advanced clinical grades (ISUP 4 and 5) each represent less than 12% of the total.

The PANDA images originate from two distinct medical centers: Radboud University Medical Center and Karolinska Institutet. According to the authors⁸, the slides from Radboud were annotated by trained students, whereas experienced pathologists

labeled those from Karolinska. This heterogeneity introduces inter-institutional variability in the annotations. In addition, the public dataset contains various forms of noise, including annotations and pen markings near tumor regions, which can adversely affect model training quality.

ISUP Grade	Karolinska	Radboud	Total
0	1,520 (27.9%)	1,372 (26.6%)	2,892 (27.24%)
1	1,404 (25.7%)	1,262 (24.4%)	2,666 (25.11%)
2	690 (12.6%)	653 (12.7%)	1,343 (12.65%)
3	635 (11.6%)	607 (11.8%)	1,242 (11.70%)
4	641 (11.7%)	608 (11.8%)	1,249 (11.77%)
5	566 (10.3%)	658 (12.7%)	1,224 (11.53%)
Total	5,456 (51.3%)	5,160 (48.7%)	10,616 (100%)

Source: Data derived from the PANDA Challenge dataset (2020).

TABLE 1 Characteristics of the PANDA dataset used in this study.

3.2 | Preprocessing

The preprocessing pipeline was structured in a sequential manner to ensure evaluation integrity and sample quality. The process begins with a stratified data split, followed by an uncertainty-based (entropy) filtering step to remove noisy instances. Finally, to address the high dimensionality of whole-slide images and to increase the variability of the training set, patch extraction and data augmentation techniques are applied.

3.2.1 | Data Partitioning and Noise Filtering

Initially, a stratified split of the dataset was performed to preserve the original class distribution. Of the total images, 15% were reserved for the final test set, while the remaining 85% comprised the development set.

Considering the presence of noisy labels, as illustrated in Figure 2, which shows an image with visible pen markings, and the high complexity involved in discriminating specific samples (Section 3.1), a filtering mechanism based on Shannon entropy was implemented. This metric, widely used in Information Theory and Artificial Intelligence applications, allows quantifying the degree of uncertainty associated with the model predictions¹⁵.

For each instance, the model generates a probability distribution over the K classes, denoted by $p = [p_1, \dots, p_K]$. The entropy is then defined as:

$$H(p) = - \sum_{i=1}^K p_i \log(p_i), \quad (1)$$

where low values indicate higher model confidence, as the probability distribution is concentrated in a few classes, in contrast, higher values are associated with more uniform distributions, reflecting greater predictive uncertainty and suggesting that the sample may be ambiguous, noisy, or of low quality. Therefore, entropy was adopted as a systematic indicator of sample difficulty and was applied exclusively to the development set.

The filtering process was conducted in three stages. First, an EfficientNet-B0 model pretrained on ImageNet¹⁶ was used to estimate the probability distributions of all samples in the development set, from which entropy values were computed. Next, the impact of removing the highest-entropy instances was evaluated by testing thresholds for excluding the top 10%, 5%, and 2% most uncertain examples. Finally, the 2% threshold was selected as it provided the best balance between noise reduction and preservation of the available data volume.

After removing the noisy samples, the development set was stratified into 80% for training and 20% for validation. Table 2 presents the data distribution before and after filtering.



FIGURE 2 Example of an image containing noise.

Subset	% of Total	No. of Images
Test	15%	1,590
Development Set (Pre-Filtering)		9,024
Development Set (Post-Filtering)		8,844
Training	$\approx 68\%$	7,075
Validation	$\approx 17\%$	1,769
Total Used	100%	10,434

TABLE 2 Dataset distribution after noise filtering (2% threshold).

3.2.2 | Patch extraction

Given the size of the WSIs and the presence of non-informative areas, a preprocessing step was applied to optimize computational resources while preserving regions relevant to diagnosis. Initially, fixed patches of 256×256 pixels were extracted, selected based on empirical criteria for mean intensity and saturation in the RGB channels, to eliminate background regions, artifacts, and areas lacking diagnostic staining. Figure 3 illustrates the original histopathological image used as the basis for patch extraction and the resulting composite image after patch extraction and rearrangement, providing a visual understanding of the process.

For each slide, 36 patches were extracted, a number defined through preliminary experiments that evaluated the impact of the number of regions on morphological representativeness and computational feasibility. To support this choice, three configurations were compared: 25, 36, and 49 patches per slide, as illustrated in Figure 4.

The extraction with 25 patches produced a compact composition, thereby generating fewer data to analyze, which may reduce computational demands during model training and validation. However, this configuration limits slide coverage, resulting in a significant loss of morphologically relevant patterns for classification. In contrast, extracting 49 patches expanded the analyzed area but increased computational and storage costs, and included many empty or redundant regions that may negatively affect processing efficiency. Moreover, including peripheral areas with low discriminative content can compromise representation quality by diluting or saturating important features.

Thus, the intermediate configuration with 36 patches provides an appropriate balance between morphological coverage and computational efficiency, capturing most of the relevant information without overloading the image with irrelevant regions. For this reason, this configuration was adopted as the standard in this work. After extraction, the patches were rearranged and concatenated into a single composite image. Unlike approaches that preserve the original spatial distribution, in this study the patches were ordered by the amount of morphological information they contained, estimated from intensity and saturation measures in the RGB channels.

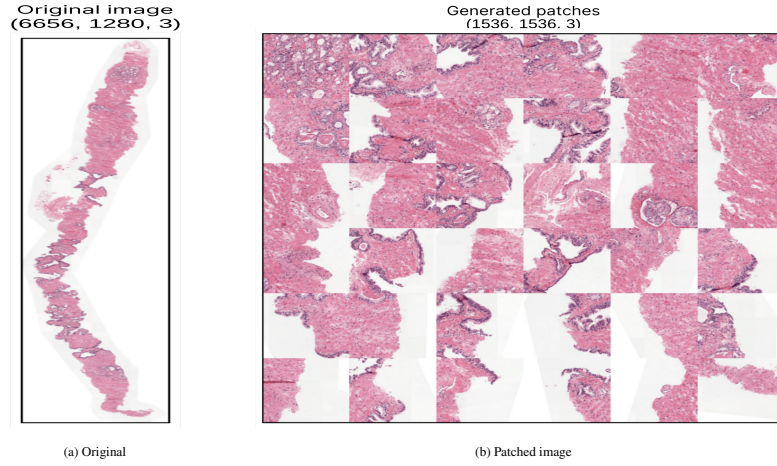


FIGURE 3 Original histopathological image (a) and composite image after patch extraction and rearrangement (b).

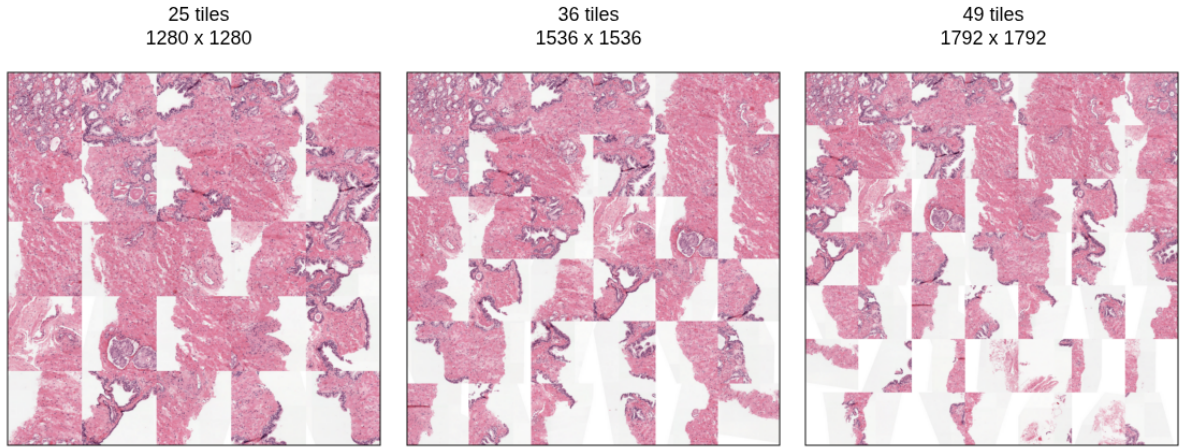


FIGURE 4 Representation of patch extraction with different quantities. The image shows three examples: (a) with 25 patches, (b) with 36 patches, and (c) with 49 patches, displayed from left to right.

3.3 | Single Classifiers

In the initial phase of the experiments, the EfficientNet-B0 model was adopted as a reference for hyperparameter optimization. A baseline was first established using the Binary Cross-Entropy (BCE) loss function. This choice was motivated by the nature of the problem, considering that a single slide may exhibit multiple lesion grades. During the baseline training, several strategies were evaluated, including Stochastic Weight Averaging (SWA), Test Time Augmentation (TTA), and variations in learning rate, number of epochs, optimizers, and color spaces. Across all experiments, the model exhibited early convergence (between epochs 7 and 10), followed by overfitting, as illustrated in Figure 5.

The main hypothesis for this behavior is that the BCE loss treats classes as independent nominal entities. Consequently, the model is penalized equally for confusing ISUP 0 with ISUP 1 (a minor clinical error) and for confusing ISUP 0 with ISUP 5 (a severe clinical error). This limitation motivated the adoption of an ordinal regression-based approach.

The proposed architecture employs ordinal regression to model the continuous and hierarchical nature of the ISUP grading system. The K -class problem is reformulated as $K - 1$ binary classification tasks, where the k -th output corresponds to the conditional probability $P(y > k)$. The target labels (y) are mapped to ordinal vectors $V = [v_0, \dots, v_{K-2}]$, formally defined as:

$$v_k = \begin{cases} 1, & \text{if } y > k \\ 0, & \text{otherwise} \end{cases}$$

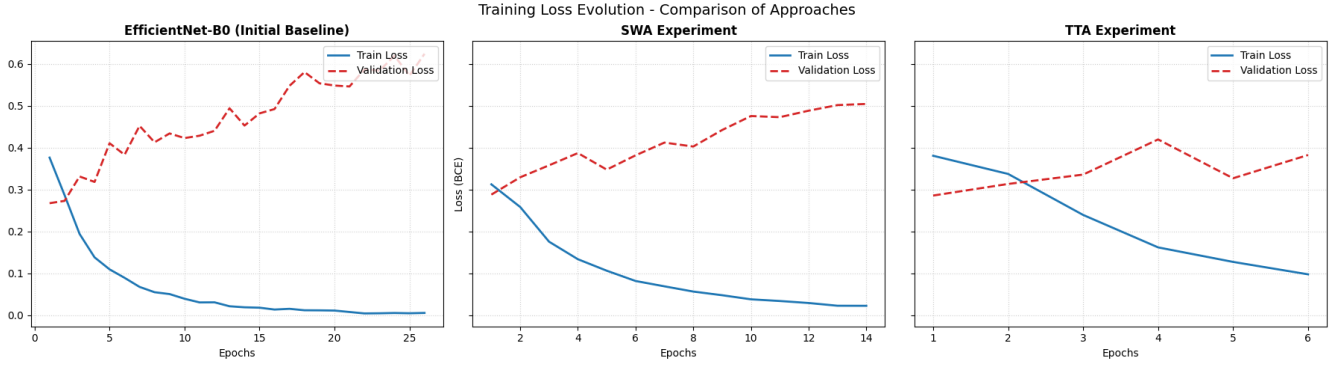


FIGURE 5 Baseline model training evolution curve.

Thus, an ISUP grade 3 is encoded as $[1, 1, 1, 0, 0]$.

This structure imposes a monotonicity constraint on the labels, preventing logically inconsistent configurations and enabling the model to capture the hierarchical relationships among cancer severity grades. Subsequently, the Focal Loss technique¹⁷ was integrated into the ordinal approach to mitigate class imbalance. Focal Loss reshapes the standard cross-entropy loss by introducing a modulation factor $(1 - p_t)^\gamma$, which reduces the relative weight of easy (well-classified) examples, thereby forcing the model to focus training on hard and misclassified samples.

Table 3 presents the hyperparameters used for training the models.

Configuration	Parameter	Value
Training	Initial learning rate	3×10^{-4}
	Maximum epochs	50
	Batch size	3
	Early stopping (patience)	7
	Warmup	1 epoch (factor 2)
	Scheduler	Cosine Annealing
	Optimizer	Adam
	Fine-tuning	150 layers
Regularization	Dropout	0.6
	Data augmentation	Flip/Transpose ($p = 0.5$)
	Entropy filtering	Top 2% removed
Focal Loss	α (alpha)	[0.25, 0.35, 0.50, 0.70, 0.90]
	γ (gamma)	2.0
Inference	Aggregation	Mean of predictions
Architectures	Tested backbones	EfficientNet-B0/B3/B7
	Output classes	6

TABLE 3 Experiment configuration for the proposed approach.

3.4 | Ensemble

To maximize predictive performance, we adopted an ensemble composed of the EfficientNet-B0, B3, and B7 models. The literature supports this approach due to its greater robustness and generalization capabilities compared to using individual models¹⁸. Different strategies were evaluated to combine the predictions.

In the simple averaging strategy, the predicted probabilities from each model are combined using an arithmetic mean, assuming that all models contribute equally to the final decision. This approach is simple and effective, especially when the models perform similarly. The weighted averaging strategy follows the same principle. Still, it assigns different weights to each model, typically

defined based on individual performance on a validation set, so that more accurate models exert greater influence on the final prediction.

In the maximum strategy, the final decision is obtained by selecting, for each class, the highest predicted probability among the ensemble models. This approach emphasizes the most confident prediction across models and can be advantageous when one model captures specific patterns with higher accuracy. Finally, in majority voting, each model produces a discrete prediction, and the class most frequently predicted is selected as the final output. This strategy helps reduce the impact of inconsistent or noisy predictions from individual models.

The comparative analysis of these strategies enabled the assessment of the trade-off among simplicity, robustness, and sensitivity to performance for each model, supporting the selection of the most appropriate approach for the problem under study.

3.5 | Evaluation Metrics

For evaluation, two metrics were analyzed: accuracy,

$$A = \frac{TP + TN}{TP + TN + FP + FN},$$

where TP (true positives) corresponds to cases in which the model correctly identifies a class, and TN (true negatives) represents instances correctly recognized as not belonging to that class. The quantities FP (false positives) and FN (false negatives) refer, respectively, to errors in which the model incorrectly assigns a class or fails to recognize a case that belongs to it. In addition, the quadratic weighted Cohen's Kappa¹⁰ was used:

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}, \quad w_{ij} = \frac{(i-j)^2}{(k-1)^2}.$$

Here, O_{ij} and E_{ij} denote the observed and expected confusion matrices, respectively, and k is the number of classes. This metric is well suited to the ordinal nature of Gleason grades, as it assigns higher penalties ($w_{ij} \propto (i-j)^2$) to errors between more distant classes.

4 | RESULTS

Nesta seção apresentamos a avaliação quantitativa dos modelos individuais e das abordagens de ensemble, seguindo a metodologia descrita na Seção 3.

4.1 | Evolution of the EfficientNet-B0 Base Model

Starting from a baseline model with a Kappa of 0.826, incremental improvements were applied, resulting in consistent performance gains. The adoption of Ordinal Loss increased the κ_{quad} from 0.838 to 0.851, highlighting its effectiveness in capturing the ordinal structure of the classes and subsequently, combining Focal Loss with entropy-based filtering further improved performance, reaching a κ_{quad} of 0.856 and establishing the best set of hyperparameters for the EfficientNet-B0 architecture, outperforming the baseline by 3.0 percentage points. Figure 6 shows that the validation loss remained stable, while the κ_{quad} steadily improved up to approximately 0.860, suggesting consistent model improvement without significant signs of overfitting.

4.2 | Impact of Model Depth

We applied the same pipeline structure (learning rate, number of epochs, optimizer, and loss functions) to deeper architectures (EfficientNet-B3 and B7). Contrary to initial expectations, the obtained results were 0.848 and 0.844, respectively, both lower than that achieved by EfficientNet-B0. This observation suggests that simply increasing model capacity, without architecture-specific hyperparameter adjustments or appropriate regularization strategies, not only failed to yield benefits but also led to

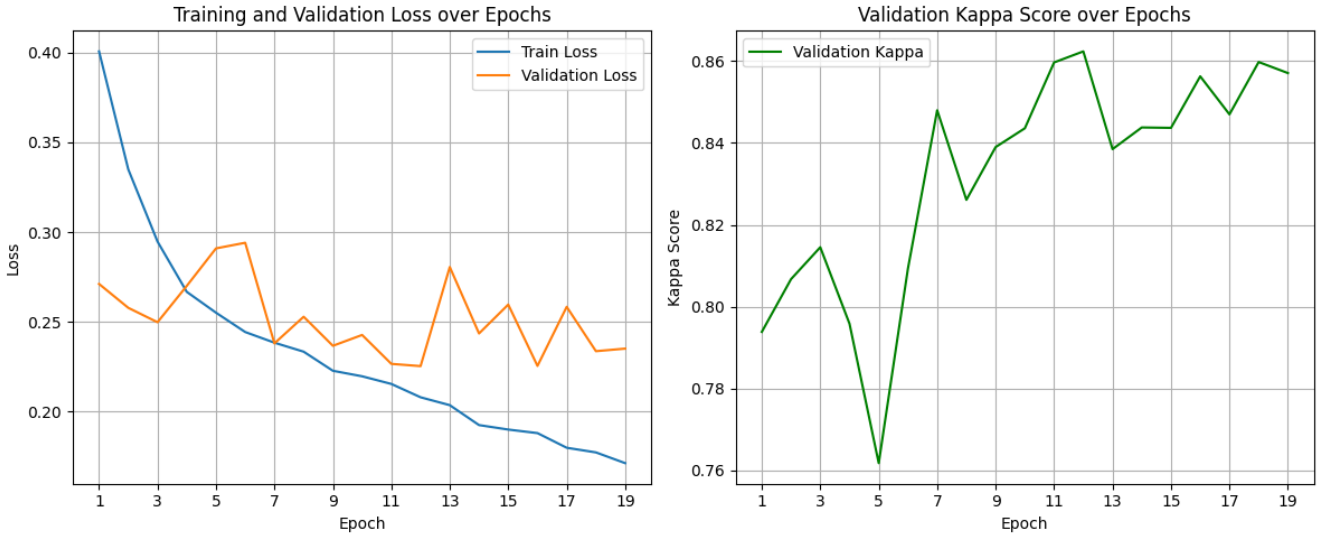


FIGURE 6 Evolution of the loss and Kappa during training of the best individual model (EfficientNet-B0 with ordinal regression, Focal Loss, and entropy-based filtering).

slight performance degradation. This behavior indicates a higher susceptibility to overfitting or the need for a more refined optimization process for deeper architectures.

4.3 | Performance of Ensemble Strategies

Table 4 presents a comparison of the methods, including QWK, accuracy, standard deviations, and 95% confidence intervals. Consistently, the ensemble strategies outperformed all individual models. The simple averaging method achieved the best performance, with $QWK = 0.879$ and an accuracy of 69.8%, representing a gain of 2.3 percentage points over the best individual model (B0-Entropy-Ordinal-Focal). This result indicates that combining predictions leverages complementarities across different architectures and enhances their robustness.

The weighted averaging ensemble achieved virtually equivalent performance ($QWK = 0.878$) to simple averaging, with only a marginal difference, suggesting that weight assignment did not yield a significant gain under the evaluated conditions, possibly due to the need for more refined calibration. The majority voting method performed poorly ($QWK = 0.859$), though it still outperformed individual models, suggesting that strategies based on discrete aggregation are less effective than probability-based combinations. The maximum-based method exhibited the worst performance ($QWK = 0.839$), even falling below several individual models, suggesting that selecting the most confident prediction tends to amplify uncertainties and introduce instability into the decision process.

In terms of accuracy, the same pattern was observed: the simple averaging ensemble reached 69.8%, while individual models ranged between 60.2% and 66.0%. These results reinforce that model aggregation improves not only ordinal agreement (QWK) but also the rate of correct classifications, highlighting the positive impact of ensemble strategies on the stability and generalization of the system.

5 | DISCUSSION

The results show that combining ordinal modeling, entropy-based filtering, and ensemble strategies consistently improves the classification of ISUP grades, even in a high-heterogeneity, noisy setting such as the PANDA dataset. The comparison between the baseline and the B0-Entropy-Ordinal-Focal model indicates that explicitly modeling the ordinal structure reduces clinically relevant errors and concentrates misclassifications between adjacent classes, as expected in diagnostic contexts.

Model	Kappa			Accuracy		
	Result	Std	CI 95%	Result	Std	CI 95%
<i>Ensemble Strategies</i>						
Ensemble-Mean	0,879	0,010	[0,858; 0,898]	0,698	0,011	[0,676; 0,721]
Ensemble-Weighted	0,878	0,009	[0,858; 0,895]	0,692	0,011	[0,669; 0,714]
Ensemble-Vote	0,859	0,011	[0,835; 0,882]	0,692	0,011	[0,668; 0,715]
Ensemble-Max	0,839	0,011	[0,815; 0,859]	0,582	0,012	[0,559; 0,605]
<i>Individual Models</i>						
B0-Entropy-Ordinal-Focal	0,856	0,011	[0,833; 0,876]	0,660	0,012	[0,635; 0,683]
B0-Entropy-Ordinal-Focal SWA	0,852	0,011	[0,830; 0,873]	0,653	0,012	[0,629; 0,676]
B3-Entropy-Ordinal-Focal	0,848	0,012	[0,824; 0,870]	0,660	0,011	[0,638; 0,683]
B7-Entropy-Ordinal-Focal	0,844	0,011	[0,821; 0,865]	0,602	0,012	[0,578; 0,628]
B0-Baseline	0,826	0,012	[0,806; 0,846]	0,592	0,012	[0,572; 0,612]

TABLE 4 Performance comparison: Kappa and Accuracy with Standard Deviation and 95% Confidence Interval.

The analysis of the confusion matrices presented in Figure 7 shows that the combination of ordinal losses with Focal Loss improves not only the global metrics but also the clinical coherence of the predictions. Compared to the baseline, which exhibited greater dispersion of errors among intermediate classes, the proposed models concentrated predictions along the main diagonal, with more pronounced improvements in ISUP grades 3 to 5, which are of higher prognostic relevance. This behavior is fundamental in clinical applications, where significant ordinal-distance errors directly impact therapeutic decision-making.

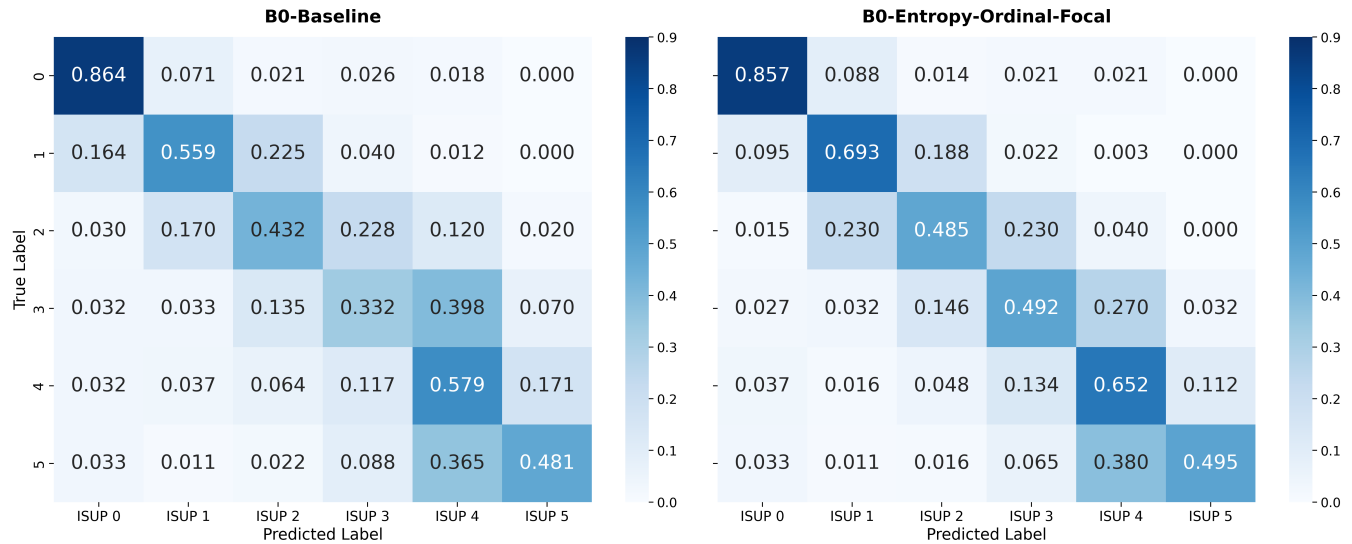


FIGURE 7 Comparison between the baseline and the best individual model

The analysis of the confusion matrix of the Ensemble model using the simple averaging strategy (Figure 8) confirms the robustness gains observed in the global metrics, with a clear concentration of predictions along the main diagonal. The model maintains high effectiveness in identifying low-risk cases (ISUP 0 and 1) and consistent performance across the intermediate grades (ISUP 2 and 3), with errors predominantly confined to adjacent classes, in line with the expected behavior of ordinal models. For the more aggressive grades, a complementary pattern relative to the individual models is observed. Although performance on ISUP 4 is slightly inferior to that of the B0-Entropy-Ordinal-Focal model, the Ensemble shows superior ability

to identify ISUP 5 cases, reinforcing its clinical utility by prioritizing sensitivity in the histological grade associated with the worst prognosis.

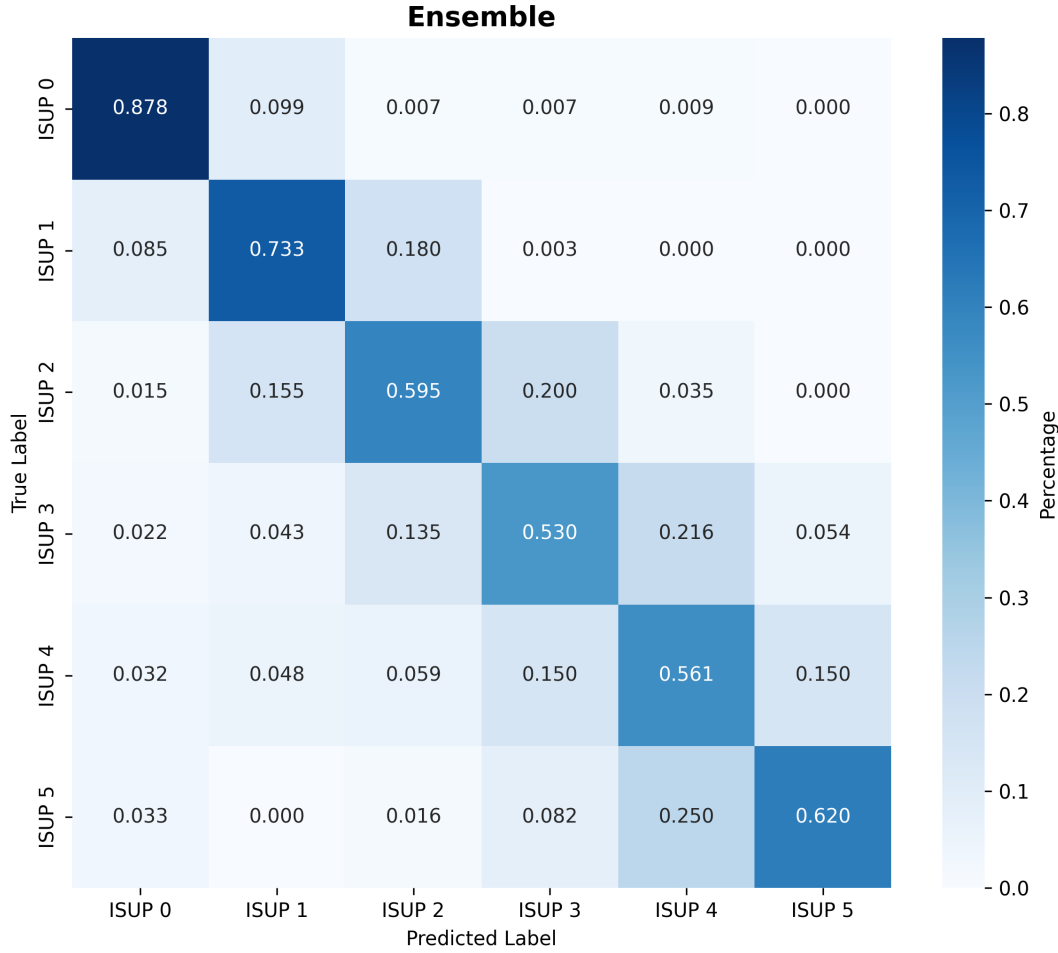


FIGURE 8 Ensemble confusion matrix.

The Table 5 presents a comparison between the results obtained in this work and those reported in previous studies. Given the methodological heterogeneity in the literature, the analysis was structured around three principal axes: data complexity, performance as measured by Quadratic Weighted Kappa (QWK), and clinical consistency.

Regarding data complexity, this work uses the PANDA dataset, widely recognized for its high levels of noise and institutional variability. In contrast, studies such as those by Lucas et al. and Hammouda et al. report high performance metrics on private, smaller-scale datasets. This scenario may overestimate performance and limit the generalizability of the models.

In terms of QWK performance, the proposed model achieves competitive results, outperforming works such as those by Lucas et al. and Nagpal et al. Although Singhal et al. reported higher values, their approach relies on deeper architectures and more complex strategies, including active learning. In this study, the EfficientNet family was selected to achieve a more balanced trade-off between predictive performance and computational cost. From a clinical perspective, although some studies report higher accuracy values, the confusion matrix analysis of the present work shows that the model more robustly preserves the ordinal structure of the problem. Most errors occur between adjacent classes, a behavior consistent with clinical practice and not adequately captured by aggregate metrics such as accuracy.

Study	Methodology	Dataset	Performance	Remarks
Proposed	EfficientNet Ens. + Ordinal Focal	PANDA (10,616 WSIs)	QWK: 0.879	Focus on noise reduction (Entropy) and ordinal consistency.
³	Extended U-Net	Radboud (5,759 biopsies)	QWK: 0.918	Outperformed 10 out of 15 pathologists.
¹⁴	U-Net + ResNeXt50	Multicenter (6,670 WSIs)	QWK: 0.930	Used <i>Active Learning</i> and domain adaptation.
⁵	InceptionV3 + k-NN	TCGA + Others (1,226)	Acc: 0.700	Significantly outperformed the average of pathologists.
¹²	InceptionV3	Private (96 sections)	QWK: 0.700	Limited dataset; high accuracy for binary classes.
¹¹	ResNet-18	Private (85 WSIs)	Acc: 85.4%	Pilot study focused on multiclass grading.
¹³	Pyramidal CNN	Radboud (608 images)	Acc: 94.0%	Small dataset, artificially balanced.

Abbreviations: QWK, Quadratic Weighted Kappa; Acc, Accuracy; WSI, Whole Slide Image.

TABLE 5 Comparison of the results obtained by the proposed method with related state-of-the-art works.

6 | CONCLUSION

This study presented an efficient and robust pipeline for the automated classification of ISUP grades in prostate histopathology images, combining ordinal regression, Focal Loss, and entropy-based filtering with variants of the EfficientNet architecture. The results demonstrated that the proposed approach substantially improves the clinical coherence of predictions, reduces severe ordinal errors, and outperforms individual models by leveraging ensembles based on probabilistic averaging.

The simple averaging ensemble achieved the best overall performance, with QWK = 0.879 and an accuracy of 69.8%, highlighting that the combination of complementary architectures provides greater stability and generalization. Beyond quantitative performance, the model exhibited more consistent behavior from a clinical perspective, with errors predominantly concentrated between adjacent classes.

The main limitation is the use of a single dataset, albeit multicentric and heterogeneous. Future evaluations on independent external datasets are necessary to confirm the method’s robustness and generalization. Furthermore, the greater difficulty observed in the intermediate grades suggests that specific strategies for these classes may represent a promising path for further improvements. Future work includes investigating Transformer-based architectures, especially Vision Transformers and hybrid CNN-Transformer models, as well as incorporating active learning methods and automatic uncertainty-based curation strategies, aiming to make the system even more robust and clinically applicable.

REFERENCES

1. Siegel RL, Giaquinto AN, Jemal A. Cancer statistics, 2024. *CA: A Cancer Journal for Clinicians*. 2024;74(1):12–49.
2. Epstein JI, Egevad L, Amin MB, Delahunt B, Srigley JR, Humphrey PA. The 2014 International Society of Urological Pathology (ISUP) Consensus Conference on Gleason Grading of Prostatic Carcinoma: Definition of Grading Patterns and Proposal for a New Grading System. *The American Journal of Surgical Pathology*. 2016;40(2):244–252.
3. Bulten W, Pinckaers H, Boven vH, et al. Automated deep-learning system for Gleason grading of prostate cancer using biopsies: a diagnostic study. *The Lancet Oncology*. 2020;21(2):233–241.
4. Campanella G, Hanna MG, Geneslaw L, et al. Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nature Medicine*. 2019;25(9):1301–1309.
5. Nagpal K, Foote D, Liu Y, et al. Development and validation of a deep learning algorithm for improving Gleason scoring of prostate cancer. *npj Digital Medicine*. 2019;2:48.
6. Tan M, Le QV. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. In: 2019; Long Beach, CA, USA:6105–6114.
7. Hu J, Shen L, Sun G. Squeeze-and-Excitation Networks. In: 2018; Salt Lake City, UT, USA:7132–7141.
8. PANDA Challenge . Prostate cANcer graDe Assessment Challenge. <https://www.kaggle.com/c/prostate-cancer-grade-assessment>; 2020. Accessed: 2024-01-01.
9. Izmailov P, Podoprikin D, Garipov T, Vetrov D, Wilson AG. Averaging Weights Leads to Wider Optima and Better Generalization. In: 2018; Monterey, CA, USA:876–885.
10. Cohen J. Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit. *Psychological Bulletin*. 1968;70(4):213–220.
11. Kott O, Linsley D, Amin A, et al. Development of a Deep Learning Algorithm for the Histopathologic Diagnosis and Gleason Grading of Prostate Cancer Biopsies: A Pilot Study. *European Urology Focus*. 2019;5(1):79–83.
12. Lucas M, Jansen I, Savci-Heijink CD, et al. Deep learning for automatic Gleason pattern classification for grade group determination of prostate biopsies. *Virchows Archiv*. 2019;475(1):77–83.
13. Hammouda K, Khalifa F, El-Melegy M, et al. A Deep Learning Pipeline for Grade Groups Classification Using Digitized Prostate Biopsy Specimens. *Sensors*. 2021;21(19):6708.

14. Singhal N, Soni S, Bonthu S, et al. A deep learning system for prostate cancer diagnosis and grading in whole slide images of core needle biopsies. *Scientific Reports*. 2022;12:3383.
15. Ali I, Khan A, Khan M, Ahmad R, Ullah I. Shannon Entropy in Artificial Intelligence and Its Applications Based on Information Theory. *Entropy*. 2023;25(2):364.
16. Russakovsky O, Deng J, Su H, et al. ImageNet Large Scale Visual Recognition Challenge. *International Journal of Computer Vision*. 2015;115(3):211–252.
17. Lin TY, Goyal P, Girshick R, He K, Dollár P. Focal loss for dense object detection. In: 2017:2980–2988.
18. Mian Q, Yang X, Zhang H, et al. Ensemble learning: A systematic review and meta-analysis. *Engineering Applications of Artificial Intelligence*. 2024;132:107812.